WEC 2026 problem description

Will you score a goal?

Predicting goal-scoring probability in football using performance match data

In modern professional sport and football in particular, real-time and post-match data analysis has become a standard analytical support tool used by most coaching staffs worldwide. One of the most important application areas of such analysis is the assessment of **player performance during a match**. Experienced coaches tend to rely on their intuition to identify which players are performing well and which are beginning to decline. While this intuition is often valuable, it is inevitably based on heuristic judgments that may overlook subtle patterns, contextual effects or interactions between multiple aspects of performance (Anderson & Sally, 2013; Memmert et al., 2017).

A data-driven approach enables a more **objective, systematic and reproducible evaluation of player performance**. Football performance can be described across multiple dimensions, including physical output (e.g. sprints, distance covered), technical actions (e.g. shots or passes), tactical context (e.g. formation and playing position), and behaviour under defensive pressure. Regardless of the analytical perspective adopted, football matches are ultimately decided by goals. Consequently, a fundamental problem in football analytics is estimating the probability that a given player will score a goal (Decroos et al., 2019).

You are therefore asked to build a **predictive model** to support coaches in better assessing a **player's likelihood of scoring a goal over the remaining duration of the game**. To facilitate this task, a base dataset has been prepared containing performance data for all players who participated in the **UEFA Under-21 European Championship held in Slovakia in 2025**. The dataset is structured to reflect the dynamic nature of a football match, with player performance observed in consecutive **15-minute intervals**, referred to as checkpoints.

The **base table** (players_quarters_final) is **ready-for-use** in your modelling, created based on selected sources mentioned below, and includes for **each player and each checkpoint** rolling performance aggregates describing activity in the most recent 15 minutes (variables prefixed with last15_) as well as cumulative aggregates summarizing activity from the start of the match up to the analysed moment (variables prefixed with cumul_). These aggregates capture, among other variables, information on shots taken and sprints performed. Importantly, the dataset also contains **a binary dependent**

variable scored_after, indicating whether the player scored a goal after the analysed interval. In addition to individual performance measures, contextual information such as playing position, team formation, home versus away status, and substitution timing is included, as these factors may influence goal-scoring opportunities independently of the player's own actions (Memmert et al., 2017).

In addition, you are provided with four **source and supplementary datasets** that include **event-level data**, providing more detailed descriptions of player behaviour. These include data on individual shot events (player_appearance_shot_limited), sprint and running actions (player_appearance_run), passing behavior (player_appearance_pass), and actions under defensive pressure (player_appearance_behaviour_under_pressure). The combination of aggregated checkpoint data and fine-grained event data allows for a comprehensive analysis that links micro-level actions with macro-level outcomes (Decroos et al., 2019).

The overarching research questions guiding this case is:

- **RQ1:** *How well player's on-field behaviour during a match can predict the probability that they will score a goal later in the game (measured by balanced accuracy, AUC)?*
- **RQ2:** *Which aspects of a player's on-field behaviour during a match determine the probability that they will score a goal later in the game?*

To gain deeper insights, several more detailed research questions can be formulated and addressed:

- **RQ3:** Is information about sprints and shots alone sufficient to accurately predict the probability of a player scoring a goal?
- **RQ4:** Does incorporating passing data or information about player behaviour under defensive pressure improve the predictive performance of goal-scoring models?
- **RQ5:** Does short-term performance (recent 15-minute activity) or cumulative performance up to a given point in the match have stronger influence on the probability of scoring a goal?
- **RQ6:** Does an increase in short-term performance intensity relative to a player's overall performance level increase the probability of scoring a goal?
- **RQ7:** Do external factors that are not directly controlled by the player influence the probability of the player scoring a goal, and if so, which of them matter most?

Your task is therefore not only to develop a predictive model, but also to **analyze, compare, and interpret** the role of individual behaviour, temporal dynamics, and contextual factors in goal scoring. The main aim is to generate insights that can complement coaching intuition and support better in-game and post-game decision-making.

Bibliography

Anderson, C., & Sally, D. (2013). *The Numbers Game: Why Everything You Know About Football Is Wrong*. Penguin Books.

Bunker, R. P., & Thabtah, F. (2019). A machine learning framework for sport result prediction. *Applied Computing and Informatics*, 15(1), 27–33.

Decroos, T., Bransen, L., Van Haaren, J., & Davis, J. (2019). Actions speak louder than goals: Valuing player actions in soccer. *Proceedings of the 25th ACM SIGKDD International Conference on Knowledge Discovery & Data Mining*, 1851–1861.

Memmert, D., Lemmink, K. A. P. M., & Sampaio, J. (2017). Current approaches to tactical performance analyses in soccer. *Sports Medicine*, 47(1), 1–10.

Main dataset for the contest

- **players_quarters_final.csv** - the table contains player-level football match data split into 15-minute segments, showing participation details, positions, formations, and substitution information for each player. It also records both 15-minutes interval-based and cumulative physical and attacking performance metrics, such as distance covered, sprints, speed, shots, and goals. In the data set the *scored_after* column is a dependent variable.

Source and supplementary datasets for potential extension of feature space

- **player_appearance_shot_limited.csv** – table contains all shots taken during matches, including: goals, shots on target, shots off target, blocked shots, penalties and own goals, but with NO outcome of the shot included – to avoid information leakage in the modelling process. Selected aggregates for last 15 minutes and cumulatively from the beginning of a player occurrence are already included in the base dataset.
- **player_appearance_run.csv** – table records high-intensity physical actions, specifically: high-speed runs (hsr) and sprints (sprint). Selected aggregates for last 15 minutes and cumulatively from the beginning of a player occurrence are already included in the base dataset.
- **player_appearance_pass.csv** – table records every pass made during matches, with information about: who made the pass, who the intended receiver was, whether the pass was accurate, when and where it happened. Data from this table was NOT used to create a base dataset.
- **player_appearance_behaviour_under_pressure.csv** – single on-ball event that occurs while a player is under defensive pressure (e.g. passes, ball carries, turnovers) captured during a match. Data from this table was NOT used to create a base dataset.